

CA 7101 User Interface Design

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CHATBOT



<http://knowledge.wharton.upenn.edu/article/rise-chatbots-time-embrace/>

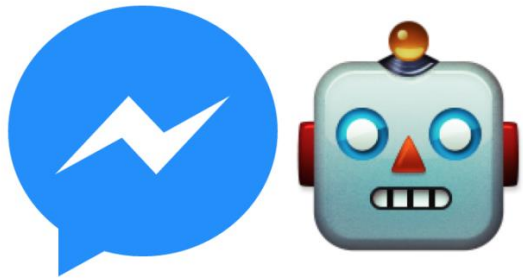
“A chatbot (also known as a talkbot, chatterbot, Bot, chatterbox, Artificial Conversational Entity) is a computer program which conducts a conversation via auditory or textual methods.”

<https://en.wikipedia.org/wiki/Chatbot>

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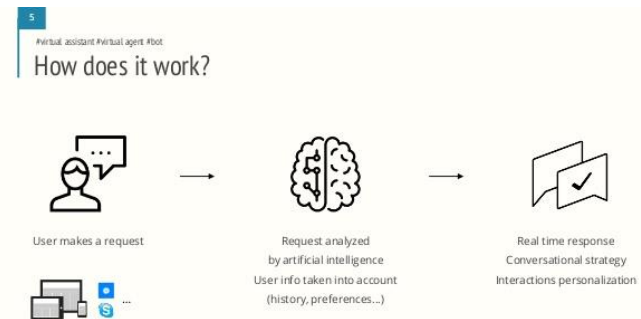
CHATBOT = CHAT + ROBOT



<http://www.garethjames.net/complete-guide-chatbots/>

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CHATBOT: HOW DOES IT WORK?



<http://www.garethjames.net/complete-guide-chatbots/>

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MACHINE LEARNING

- “Machine learning is the subfield of computer science that gives computers the ability to learn without being explicitly programmed” (Arthur Samuel, 1959).
- “Evolved from the study of pattern recognition and computational learning theory in artificial intelligence,[2] machine learning explores the study and construction of algorithms that can learn from and make predictions on data” Ron Kohavi; Foster Provost (1998). "Glossary of terms". *Machine Learning*. 30: 271-274.

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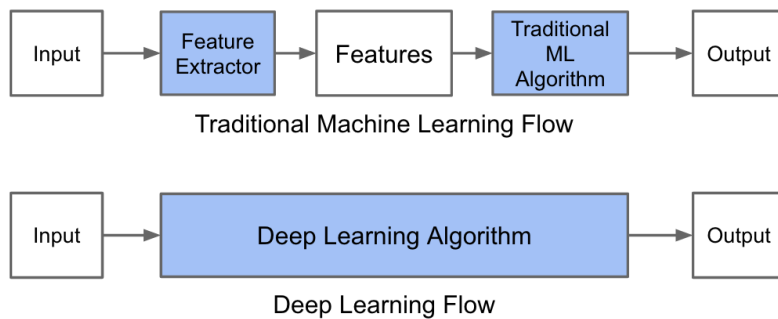
DEEP LEARNING

- “Deep learning (also known as deep structured learning, hierarchical learning or deep machine learning) is a branch of machine learning based on a set of algorithms that attempt to model high level abstractions in data.”
- “Various deep learning architectures such as deep neural networks have been applied to fields like computer vision, automatic speech recognition, and natural language processing”

https://en.wikipedia.org/wiki/Deep_learning

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MACHINE LEARNING VS DEEP LEARNING



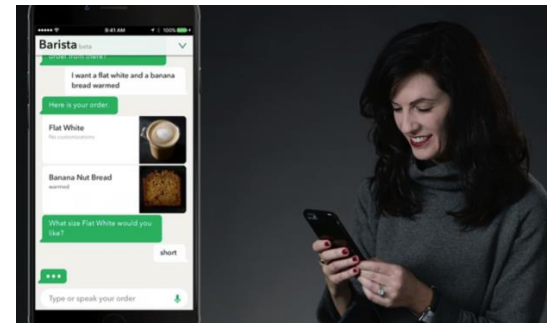
<http://adilmoujahid.com/posts/2016/06/introduction-deep-learning-python-caffe/>

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STARBUCKS UNVEILS CHATBOT APP (1)

- A voice-activated chatbot, called MyStarbucks Barista



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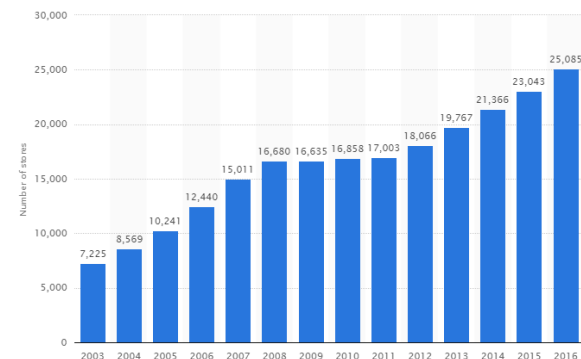
STARBUCKS UNVEILS CHATBOT APP (2)

- Let customers place orders by speaking, "I want a latte." The features can also be used to pay for orders.
- MyStarbucks Barista is only available to 1,000 people before a nationwide release this summer.

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STARBUCKS UNVEILS CHATBOT APP (3)

Number of Starbucks stores worldwide from 2003 to 2016



<https://www.statista.com/statistics/266465/number-of-starbucks-stores-worldwide/>

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MELODY (1)



<http://www.cnbc.com/2016/10/11/chinese-baidu-unveils-ai-health-chatbot-for-patients-and-doctors.html>

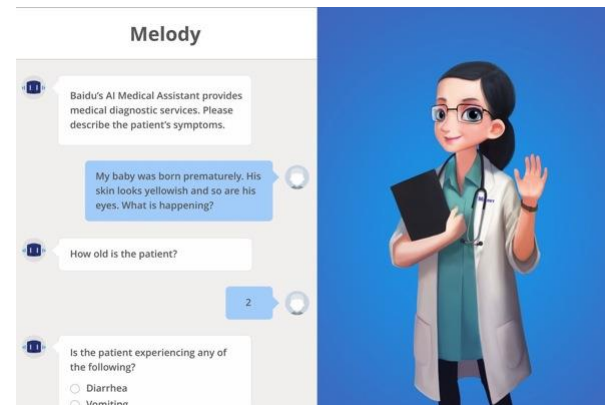
- Baidu, China's biggest search engine
- Launches medical chatbot to help Chinese doctors diagnose patients

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- Built into the company's iOS and Android Baidu Doctor app, which launched in China in 2015

MELODY (2)



<http://venturebeat.com/2016/10/11/baidu-launches-medical-assistant-chatbot-to-help-doctors-collect-patient-information/>

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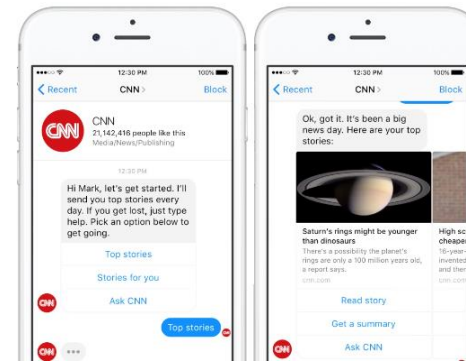
BOTS FOR MESSENGER: CNN (1)

- “Facebook announced at its F8 conference that it has created a platform for companies to develop the robots that mimic human conversation to run within its Messenger app”
- “Bots for Messenger enables users to get everything from weather updates to shopping notifications and personalized news as well as message companies and publishers as they do their friends.”

<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

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BOTS FOR MESSENGER: CNN (2)

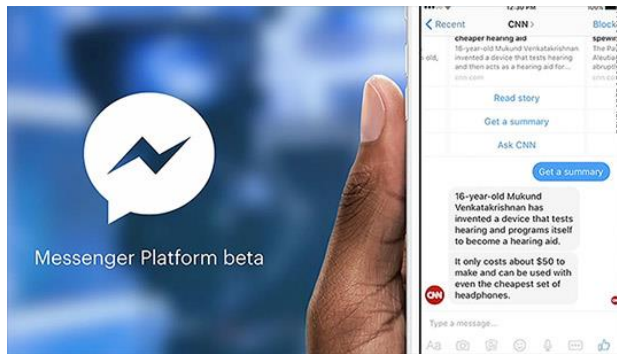


- “Launch partners include shopping app Spring, HP and CNN.”

<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

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BOTS FOR MESSENGER: CNN (3)



<http://money.cnn.com/2016/04/12/technology/facebook-messenger-bots/>

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DEEPDREAM

- DeepDream is a computer vision program created by Google
- Using a convolutional neural network to find and enhance patterns in images via algorithmic pareidolia
- Creating a dream-like hallucinogenic appearance in the deliberately over-processed images

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INTEGRATED WORK

- Deep Dream
- GANpaint

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QUESTIONS?

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